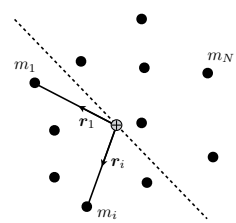
MAE 240: Space Flight Mechanics — Spring 2026

Homework 2 — Due: **2025.04.28**

1. The equations of motion for the restricted $(N+1)^{\text{st}}$-body problem, in "barycentric" form, can be stated as

$$\begin{aligned}\ddot{\boldsymbol{r}}_{N+1} &= \frac{\partial V_{\text{Bary}}}{\partial \boldsymbol{r}_{N+1}},\ V_{\text{Bary}} = \sum_{\substack{j=1\\ j\neq i}}^{N} -\frac{Gm_j}{r_{j,N+1}} \\ &= \sum_{\substack{j=1\\ j\neq i}}^{N} -\frac{Gm_j}{r_{j,N+1}^3}\boldsymbol{r}_{j,N+1},\end{aligned} \tag{1}$$

where $G = 6.6742 \times 10^{-11}$ Nm^{2}kg^{-2} is the universal gravitational constant, $m_{N+1} \ll m_i$, $i = 1, \ldots, N$, $\boldsymbol{r}_{ji} = \boldsymbol{r}_i - \boldsymbol{r}_j$, and all motion is relative to the system barycenter, shown below:



The motion $(\boldsymbol{r}_{N+1}, \dot{\boldsymbol{r}}_{N+1})$ of the $(N+1)^{\text{st}}$ body is a function of the N-body problem

$$\ddot{\boldsymbol{r}}_i = \sum_{\substack{j=1\\ j\neq i}}^{N} -\frac{Gm_j}{r_{ji}^3}\boldsymbol{r}_{ji}, \qquad i = 1, 2, \ldots, N, \tag{2}$$

but does not influence the motion $(\boldsymbol{r}_i, \dot{\boldsymbol{r}}_i, i = 1, 2, \ldots, N)$ of the N bodies.

(a) Write a generic "N-body integrator" to solve for the motion of the N bodies in the barycentric frame. Demonstrate visually that your code is working and that it maintains integration accuracy using the energy and angular momentum integrals:

$$E = T - U,\ T = \frac{1}{2}\sum_{i=1}^{N} m_i \dot{\boldsymbol{r}}_i \cdot \dot{\boldsymbol{r}}_i,\ U = \frac{1}{2}\sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{Gm_i m_j}{r_{ji}}, \tag{3}$$

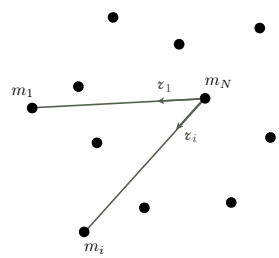
$$\boldsymbol{H} = \sum_{i=1}^{N} m_i \boldsymbol{r}_i \times \dot{\boldsymbol{r}}_i. \tag{4}$$

(b) Using the gravitational field just established, write a generic numerical simulation of the restricted $(N+1)^{\text{st}}$-body barycentric problem and show that the energy and angular momentum are not conserved.

2. The equations of motion for the restricted $(N+1)^{\text{st}}$-body problem, in "bodycentric" form, can be stated as

$$\begin{aligned}\ddot{\boldsymbol{\mathfrak{r}}}_{N+1} &= \frac{\partial V_{\text{Body}}}{\partial \boldsymbol{\mathfrak{r}}_{N+1}},\ V_{\text{Body}} = \frac{Gm_N}{\mathfrak{r}_{N+1}} + \sum_{j=1}^{N-1} Gm_j \left[\frac{1}{\mathfrak{r}_{j,N+1}} - \frac{\boldsymbol{\mathfrak{r}}_j \cdot \boldsymbol{\mathfrak{r}}_{N+1}}{\mathfrak{r}_j^3}\right] \\ &= -\frac{Gm_N}{\mathfrak{r}_{N+1}^3}\boldsymbol{\mathfrak{r}}_{N+1} - \sum_{j=1}^{N-1} Gm_j \left[\frac{1}{\mathfrak{r}_{j,N+1}^3}\boldsymbol{\mathfrak{r}}_{j,N+1} + \frac{1}{\mathfrak{r}_j^3}\boldsymbol{\mathfrak{r}}_j\right],\end{aligned} \tag{5}$$

where $\boldsymbol{\mathfrak{r}}_{ji} = \boldsymbol{\mathfrak{r}}_i - \boldsymbol{\mathfrak{r}}_j$, and all motion is relative to the m_N, shown below:



The motion $(\boldsymbol{\mathfrak{r}}_{N+1}, \dot{\boldsymbol{\mathfrak{r}}}_{N+1})$ of the $(N+1)^{\text{st}}$ body is a function of the N-body problem, restated in the bodycentric frame, as

$$\ddot{\boldsymbol{\mathfrak{r}}}_i = -\frac{G\left(m_N + m_i\right)}{\mathfrak{r}_i^3}\boldsymbol{\mathfrak{r}}_i - \sum_{\substack{j=1\\ j\neq i}}^{N-1} Gm_j \left[\frac{1}{\mathfrak{r}_{ji}^3}\boldsymbol{\mathfrak{r}}_{ji} + \frac{1}{\mathfrak{r}_j^3}\boldsymbol{\mathfrak{r}}_j\right], \qquad i = 1, \ldots, N \tag{6}$$

(a) Write a generic "N-body integrator" to solve for the motion of the N bodies bodycentric frame. Demonstrate visually that your code is working and that it maintains integration accuracy by comparing to the previous results obtained from the barycentric formulation.

(b) Using the gravitational field just established, write a numerical simulation of the restricted $(N+1)^{\text{st}}$-body bodycentric problem and compare the results to those obtained from the barycentric formalism.

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3. In the following, we specialize the N-body problem equations of motion to the two-body problem (2BP) and explore its fundamental integrals.

(a) For the case $N = 2$, write out the barycentric (Eq. 2) equations of motion for $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ and then express the motion in terms of $\boldsymbol{r} = \boldsymbol{r}_{21}$. Show that this equivalent to the bodycentric (Eq. 6) equations of motion for $\boldsymbol{\mathfrak{r}}_1 = \boldsymbol{r}_{21}$ (the central body is taken to be m_2).

(b) Define the (specific) energy and angular momentum as

$$E = \frac{1}{2}\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} - \frac{\mu}{r}, \tag{7}$$

$$\boldsymbol{H} = \boldsymbol{r} \times \dot{\boldsymbol{r}}, \tag{8}$$

where $\mu = G(m_1 + m_2)$, and define the eccentricity vector $\boldsymbol{e}$ as

$$\boldsymbol{e} = \frac{1}{\mu}\boldsymbol{v} \times \boldsymbol{H} - \frac{\boldsymbol{r}}{r}. \tag{9}$$

Show that these are constants (integrals) of the motion.

(c) Show that the integrals $(E, \boldsymbol{H}, \boldsymbol{e})$ are equivalent to only five independent conserved quantities by proving that

$$\boldsymbol{e} \cdot \boldsymbol{H} = 0, \qquad e^2 = 1 + \frac{2EH^2}{\mu^2}. \tag{10}$$

Hint: *Recall that:*

$$\begin{aligned}(\boldsymbol{a} \times \boldsymbol{b}) \cdot (\boldsymbol{c} \times \boldsymbol{d}) &= (\boldsymbol{a} \cdot \boldsymbol{c})(\boldsymbol{b} \cdot \boldsymbol{d}) - (\boldsymbol{a} \cdot \boldsymbol{d})(\boldsymbol{b} \cdot \boldsymbol{c}), \\ (\boldsymbol{a} \times \boldsymbol{b}) \cdot (\boldsymbol{a} \times \boldsymbol{b}) &= |\boldsymbol{a} \times \boldsymbol{b}|^2 \\ &= |\boldsymbol{a}|^2 |\boldsymbol{b}|^2 - (\boldsymbol{a} \cdot \boldsymbol{b})^2,\end{aligned}$$

for vectors $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}$.